Cloud Computing

Lecture 2

Data Center, Cloud Classifications and Its Characteristics

Lecture Outline

Discussion On

- Cloud Infrastructure
- Service Models of Cloud Computing
- The Cloud Stack
- Types of Clouds and CMUQ's Private Cloud
- Software Service Models

BUT FIRST OF ALL ...

WHAT IS CLOUD COMPUTING?

ROADMAP

1961: John McCarthy proposes computing as a utility
1961: IBM Services Bureau
1975: First inter-industry EDI standards (Electronic Data Interchange)
1981: SMTP defines the standard electronic mail service
1985: United Nations sponsors EDIFACT
1990: Berners-Lee invents the World-Wide Web
1994: CommerceNet
1998: RosettaNet
1999: i-Mode mobile internet
2000: IBM BCRS
2000: UDDI 1.0; "SaaS" coined
2001: Dot com bubble bursts
2005: IBM AoD
2006: Amazon EC2
2007: Google Health; force.com launch
2008: IBM ww Cloud Computing centers

1960-1970 | 1980 | 1990 | 2000 | 2010

IBM Service Bureau (1961)

Cloud computing is an evolution – rather than a revolution – the culmination of a long series of approaches to simplified IT service delivery

WHAT IS CLOUD COMPUTING

- There are many definitions of cloud computing …

- BUT there is not yet a consensus for what exactly this term means

  … is the use of computing resources (hardware and software) that are delivered as a service over a network (typically the Internet). – Wikipedia

  … refers to a variety of services available over the Internet that deliver compute functionality on the service provider's infrastructure. – Microsoft

  … is a style of computing where massively scalable IT-enabled capabilities are provided "as a service" over the network. – IBM

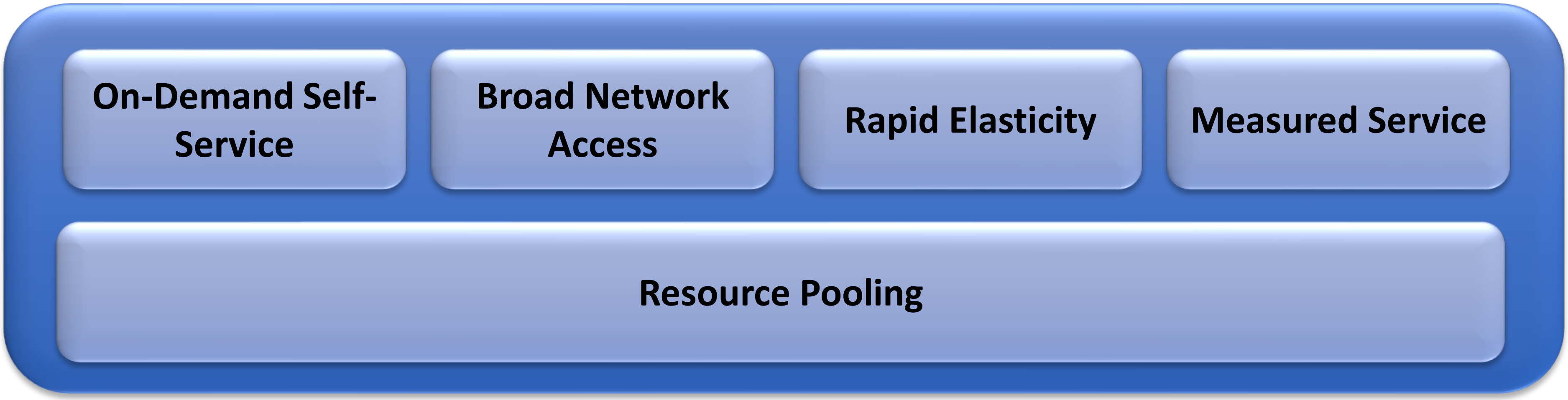
WHAT IS CLOUD COMPUTING?

- NIST definition:

  ~ everywhere

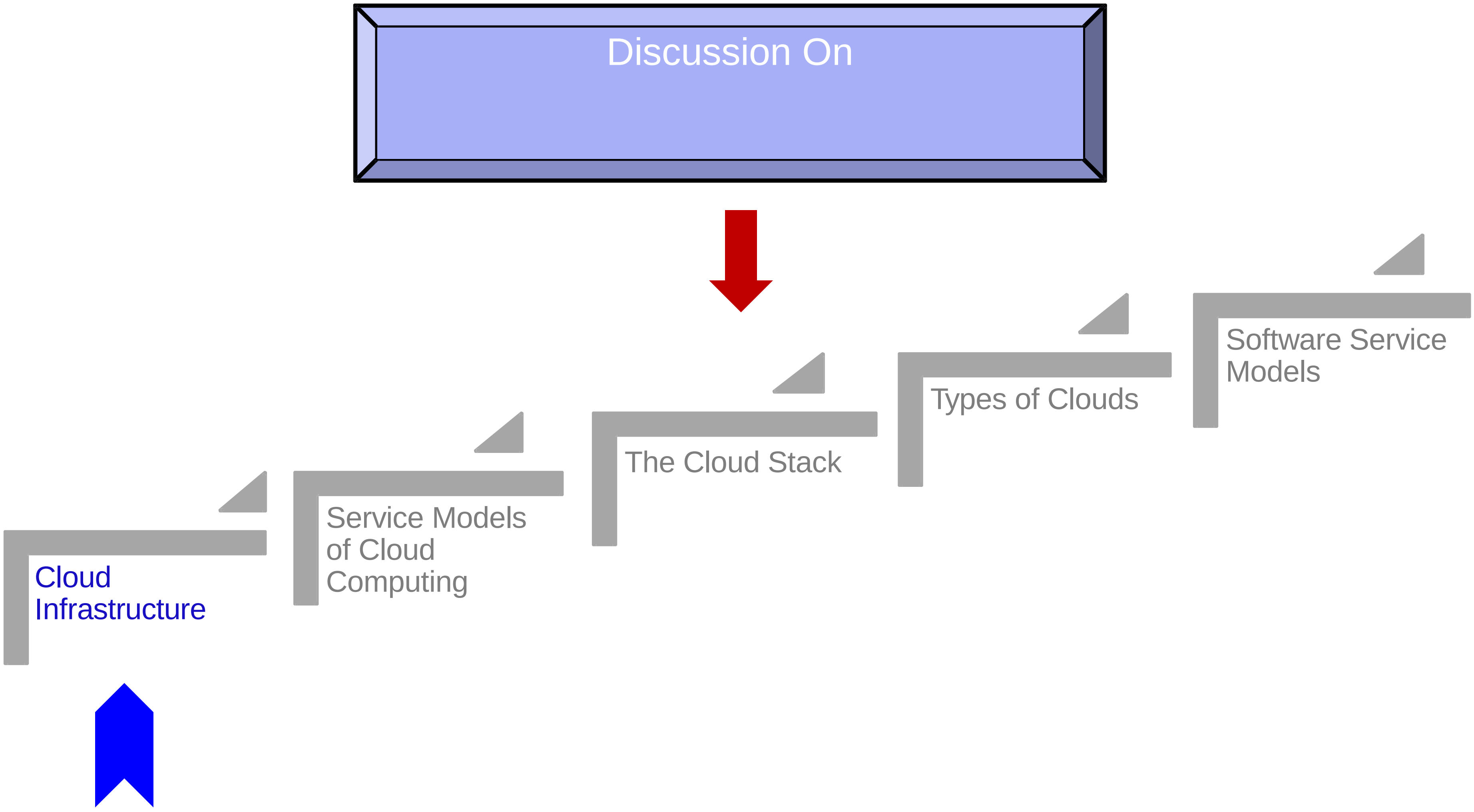
  Cloud computing is a model for enabling **ubiquitous**, convenient, **on-demand** network access to a **shared pool** of configurable computing **resources** (e.g., networks, servers, storage, applications, and services) that can be **rapidly** provisioned and released with minimal management effort or service provider **interaction**.

FIVE ESSENTIAL CHARACTERISTICS



- Cloud foundation: **Resource Pooling** (physical servers, datacenter)
  - **On-Demand Self Service**: resources can be acquired without interaction with providers
  - **Broad Network Access**: resources can be accessed over a network using different devices
  - **Rapid Elasticity**: users quickly acquire more resources by scaling out and vice versa
  - **Measured Service**: resources usage is monitored and reported

Lecture Outline



How many users and objects?

- Flickr has >6 billion photos
- Facebook has 1.15 billion active users
- Google is serving >1.2 billion queries/day on more than 27 billion items
- >2 billion videos/day watched on YouTube

How much data?

- Modern applications use massive data:
  - Rendering 'Avatar' movie required >1 petabyte of storage
  - eBay has >6.5 petabytes of user data
  - CERN's LHC will produce about 15 petabytes of data per year
  - In 2008, Google processed 20 petabytes per day
  - German Climate computing center dimensioned for 60 petabytes of climate data
  - Google now designing for 1 exabyte of storage
  - NSA Utah Data Center is said to have 5 zettabyte (!)
- How much is a zettabyte?
  - 1,000,000,000,000,000,000,000 bytes
  - A stack of 1TB hard disks that is 25,400 km high

How much computation?

- No single computer can process that much data
  - Need many computers!
- How many computers do modern services need?
  - Facebook is thought to have more than 60,000 servers
  - 1&1 Internet has over 70,000 servers
  - Akamai has 95,000 servers in 71 countries
  - Intel has ~100,000 servers in 97 data centers
  - Microsoft reportedly had at least 200,000 servers in 2008
  - Google is thought to have more than 1 million servers, is planning for 10 million (according to Jeff Dean)

Why should I care?

- Suppose you want to build the next Google
- How do you...
  - ... download and store billions of web pages and images?
  - ... quickly find the pages that contain a given set of terms?
  - ... find the pages that are most relevant to a given search?
  - ... answer 1.2 billion queries of this type <u>every day</u>?

- Suppose you want to build the next Facebook
- How do you...
  - ... store the profiles of over 500 million users?
  - ... avoid losing any of them?
  - ... find out which users might want to be friends?

- Stay tuned!

Plan for today

- Computing at scale
  - The need for scalability; scale of current services
  - Scaling up: From PCs to data centers

    NEXT
  - Problems with 'classical' scaling techniques

Scaling up

PC

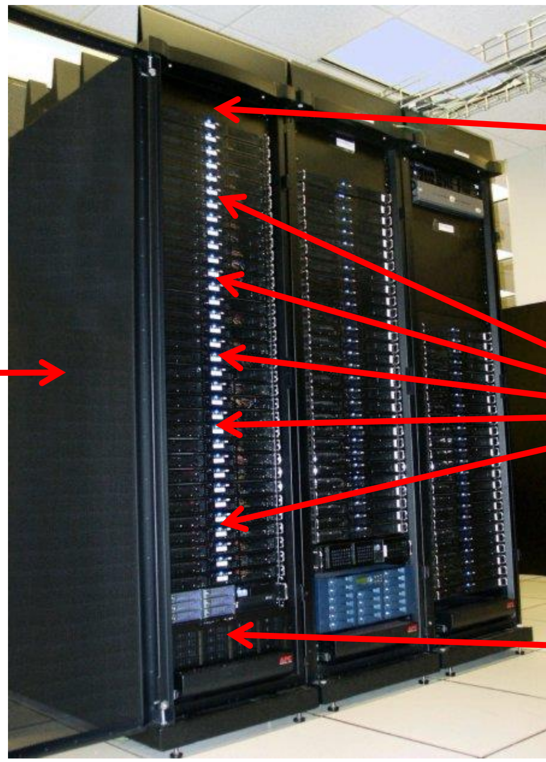
Server

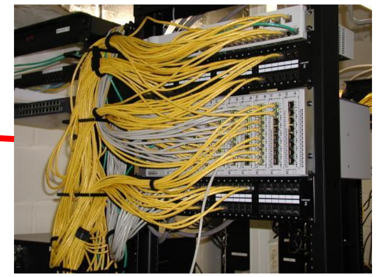
Cluster

- What if one computer is not enough?
  - Buy a bigger (server-class) computer
- What if the biggest computer is not enough?
  - Buy many computers

Clusters

Rack





Network switch (connects nodes with each other and with other racks)

Many nodes/blades (often identical)

Storage device(s)

- Characteristics of a cluster:
  - Many similar machines, close interconnection (same room?)
  - Often special, standardized hardware (racks, blades)
  - Usually owned and used by a single organization

Power and cooling

- Clusters need lots of power
  - Example: 140 Watts per server
  - Rack with 32 servers: 4.5kW (needs special power supply!)
  - Most of this power is converted into heat

- Large clusters need massive cooling
  - 4.5kW is about 3 space heaters
  - And that's just one rack!

Scaling up

PC

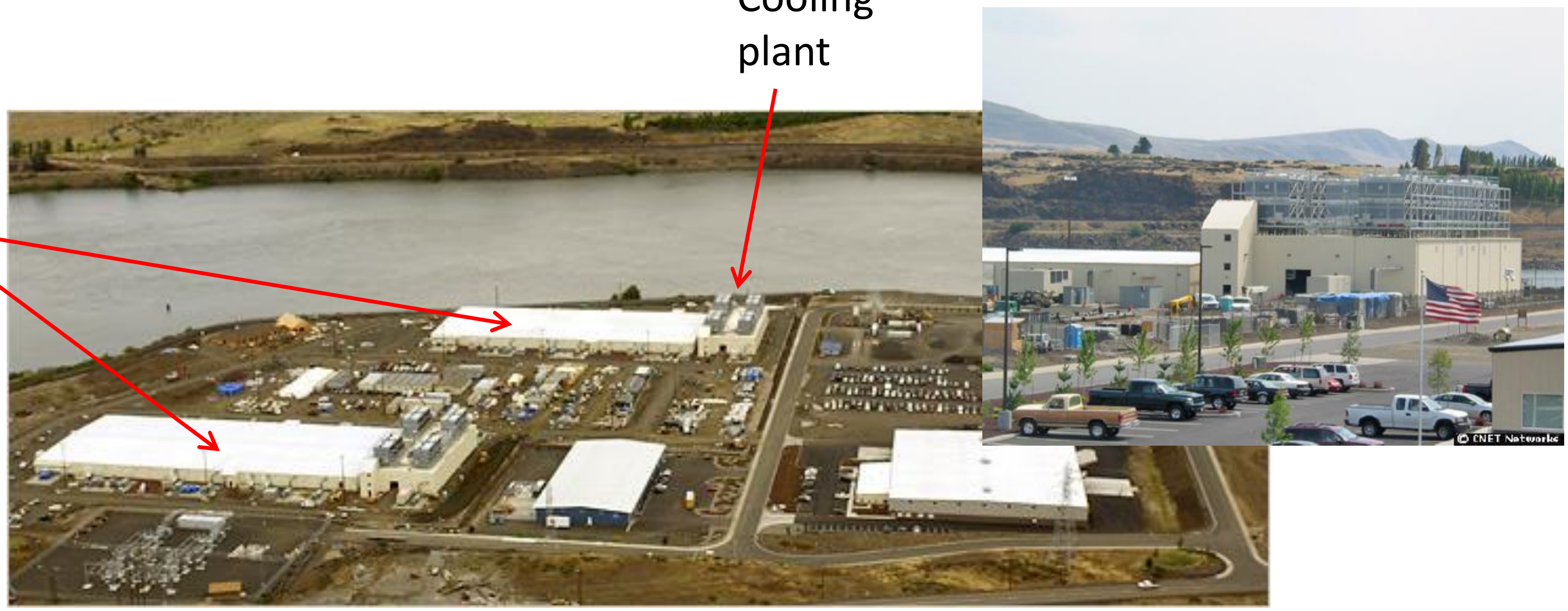
Server

Cluster

Data center

- What if your cluster is too big (hot, power hungry) to fit into your office building?
  - Build a separate building for the cluster
  - Building can have lots of cooling and power
  - Result: Data center

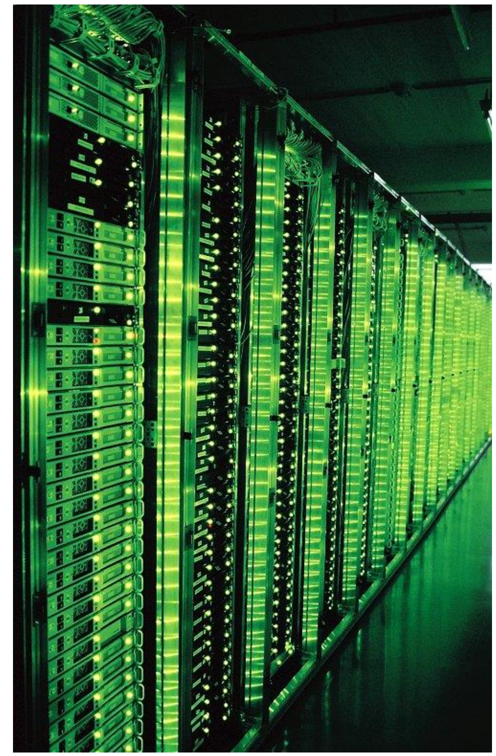
What does a data center look like?



Google data center in The Dalles, Oregon

- A warehouse-sized computer
  - A single data center can easily contain 10,000 racks with 100 cores in each rack (1,000,000 cores total)

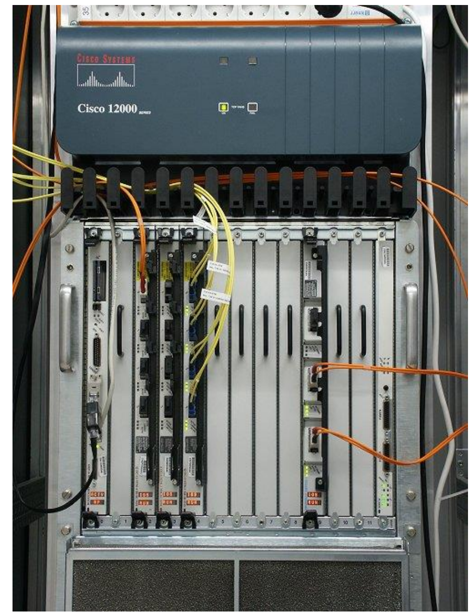
What's in a data center?



Source: 1&1

- Hundreds or thousands of racks

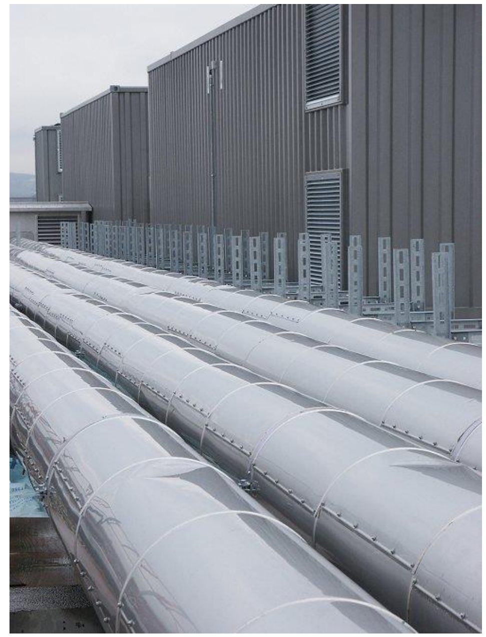
What's in a data center?



Source: 1&1

- Massive networking

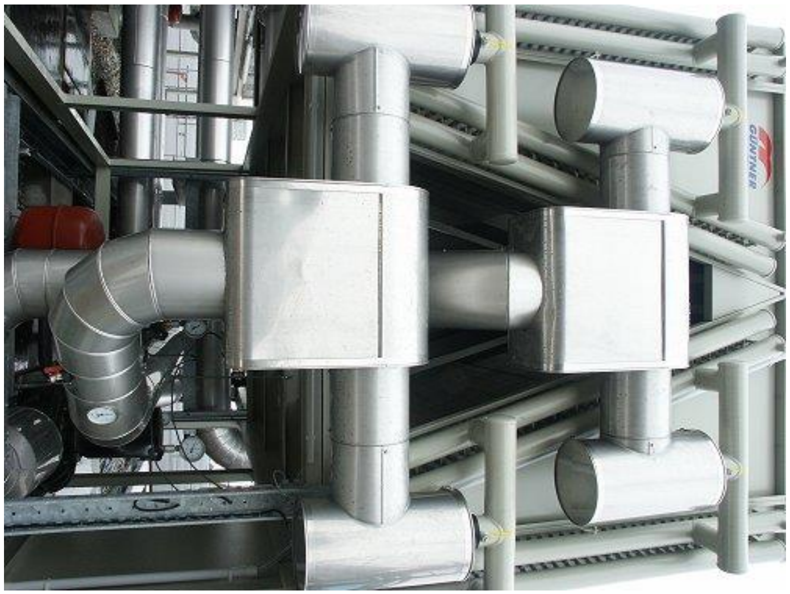
What's in a data center?


Source: 1&1

- Emergency power supplies

What's in a data center?




Source: 1&1

- Massive cooling

Energy matters!

Company	Servers	Electricity	Cost
eBay	16K	~0.6*10^5 MWh	~$3.7M/yr
Akamai	40K	~1.7*10^5 MWh	~$10M/yr
Rackspace	50K	~2*10^5 MWh	~$12M/yr
Microsoft	>200K	>6*10^5 MWh	>$36M/yr
Google	>500K	>6.3*10^5 MWh	>$38M/yr
USA (2006)	10.9M	610*10^5 MWh	$4.5B/yr

Source: Qureshi et al., SIGCOMM 2009

- Data centers consume a lot of energy
  - Makes sense to build them near sources of cheap electricity
  - Example: Price per KWh is 3.6ct in Idaho (near hydroelectric power), 10ct in California (long distance transmission), 18ct in Hawaii (must ship fuel)
  - Most of this is converted into heat → Cooling is a big issue!

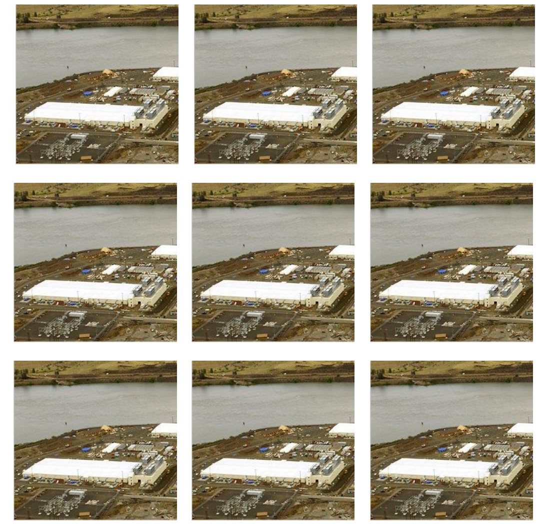
Scaling up

PC

Server

Cluster

Data center



Network of data centers

- What if even a data center is not big enough?
  - Build additional data centers
  - Where? How many?

Global distribution

- Data centers are often globally distributed
  - Example above: Google data center locations
- Why?
  - Need to be close to users (physics!)
  - Cheaper resources
  - Protection against failures

Trend: Modular data center

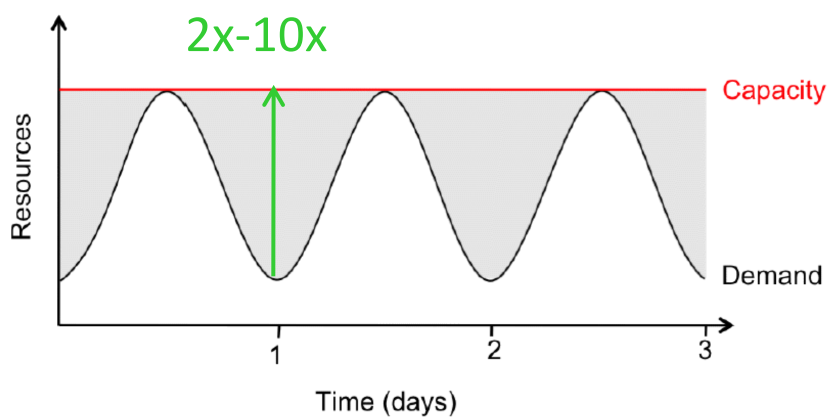
- Need more capacity? Just deploy another container!

Plan for today

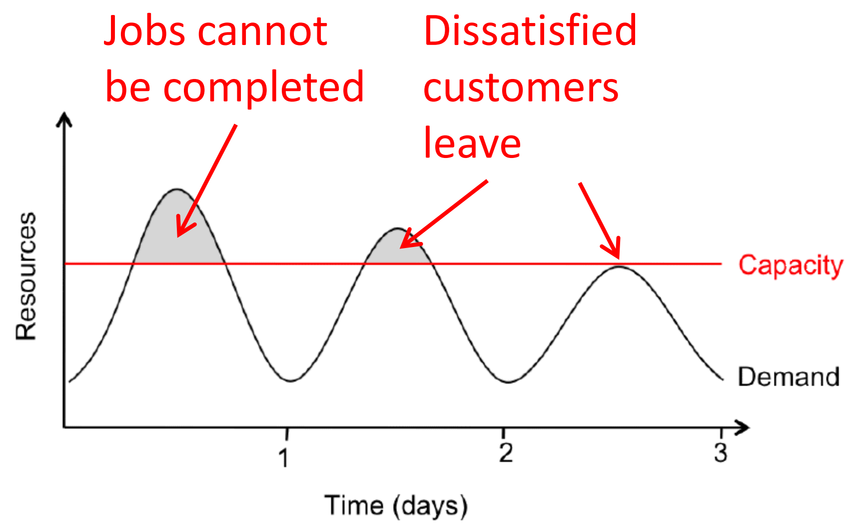
- Computing at scale
  - The need for scalability; scale of current services
  - Scaling up: From PCs to data centers
  - Problems with 'classical' scaling techniques

NEXT

Problem #1: Difficult to dimension



Provisioning for the peak load



Provisioning below the peak

- Problem: Load can vary considerably
  - Peak load can exceed average load by factor 2x-10x [Why?]
  - But: Few providers deliberately provision for less than the peak
  - Result: Server utilization in existing data centers ~5%-20%!!
  - Dilemma: Waste resources or lose customers!

Problem #2: Expensive

- Need to invest many $$$ in hardware
  - Even a small cluster can easily cost $100,000
  - Microsoft recently invested $499 million in a single data center
- Need expertise
  - Planning and setting up a large cluster is highly nontrivial
  - Cluster may require special software, etc.
- Need maintenance
  - Someone needs to replace faulty hardware, install software upgrades, maintain user accounts, ...

Problem #3: Difficult to scale

- Scaling up is difficult
  - Need to order new machines, install them, integrate with existing cluster - can take weeks
  - Large scaling factors may require major redesign, e.g., new storage system, new interconnect, new building (!)

- Scaling down is difficult
  - What to do with superfluous hardware?
  - Server idle power is about 60% of peak $\rightarrow$ Energy is consumed even when no work is being done
  - Many fixed costs, such as construction

Recap: Computing at scale

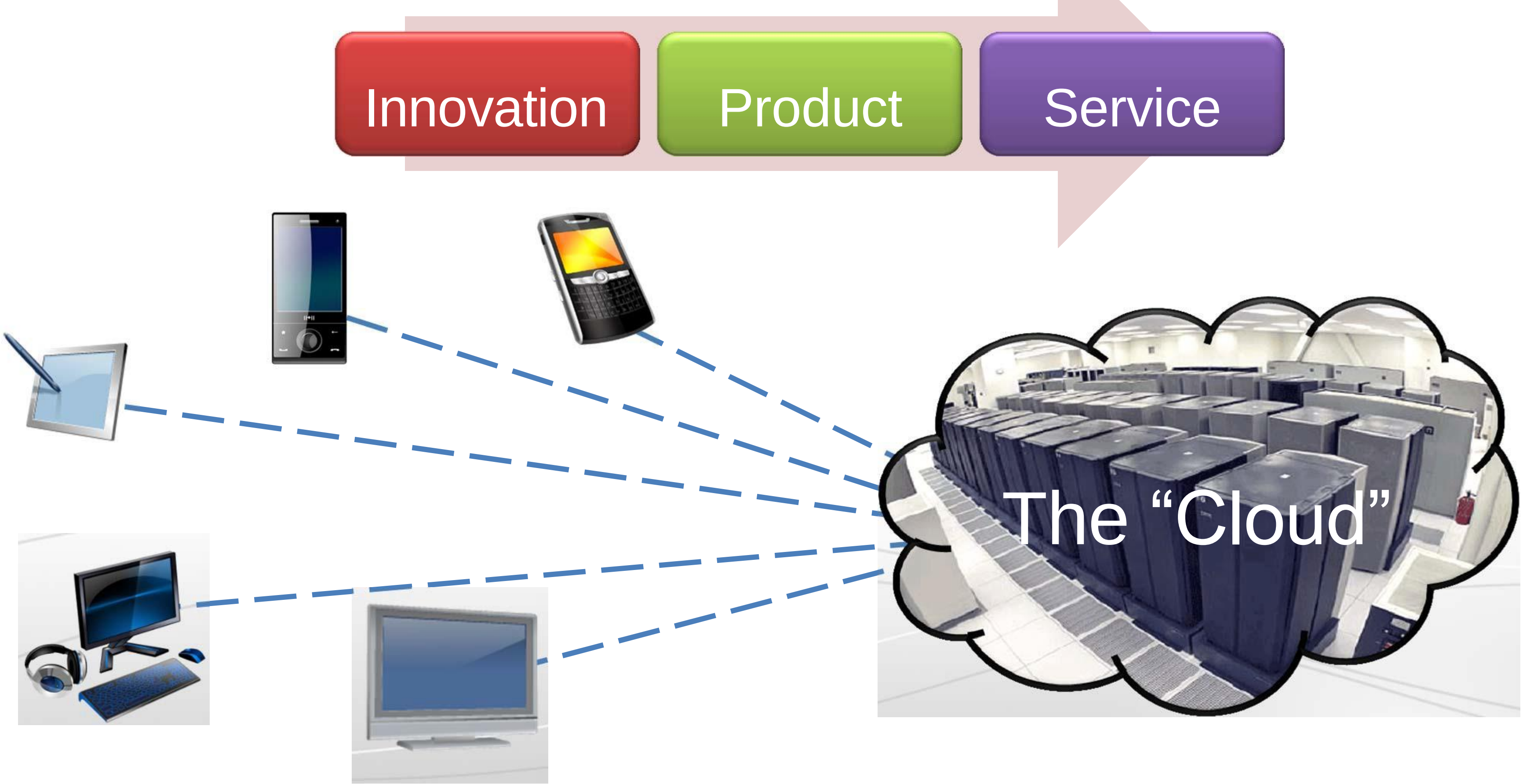
- Modern applications require huge amounts of processing and data
  - Measured in petabytes, millions of users, billions of objects
  - Need special hardware, algorithms, tools to work at this scale
- Clusters and data centers can provide the resources we need
  - Main difference: Scale (room-sized vs. building-sized)
  - Special hardware; power and cooling are big concerns
- Clusters and data centers are not perfect
  - Difficult to dimension; expensive; difficult to scale

A Cloud is …

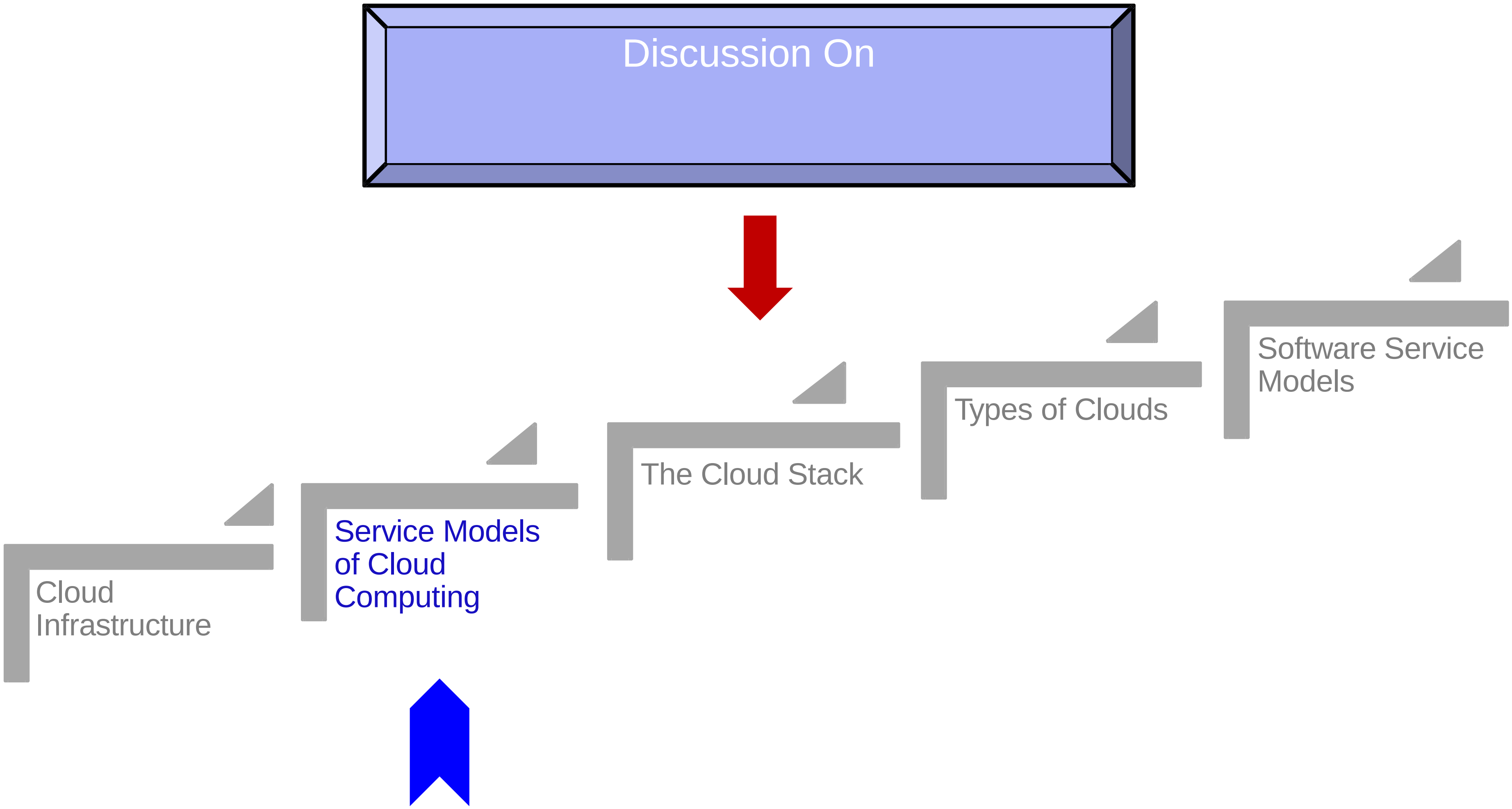
- A data center hardware and software that the vendors use to offer the computing resources and services

Cloud Computing

"Cloud Computing is the transformation of IT from a product to a service"



Lecture Outline



IT as a Service

- How do you offer IT as a service?
- Different users have different needs
- Consider the needs of:
  - Average End User
  - Mobile Application Developer
  - Enterprise System Architect

Let us look at some of the typical service models

Cloud Service Models

SaaS

- **Software-as-a-Service**
  - Applications running on browsers

PaaS

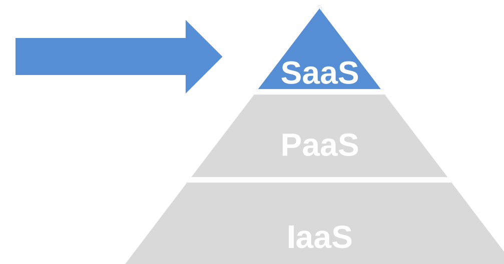
- **Platform-as-a-Service**
  - A software platform that is made available to developers to build cloud applications

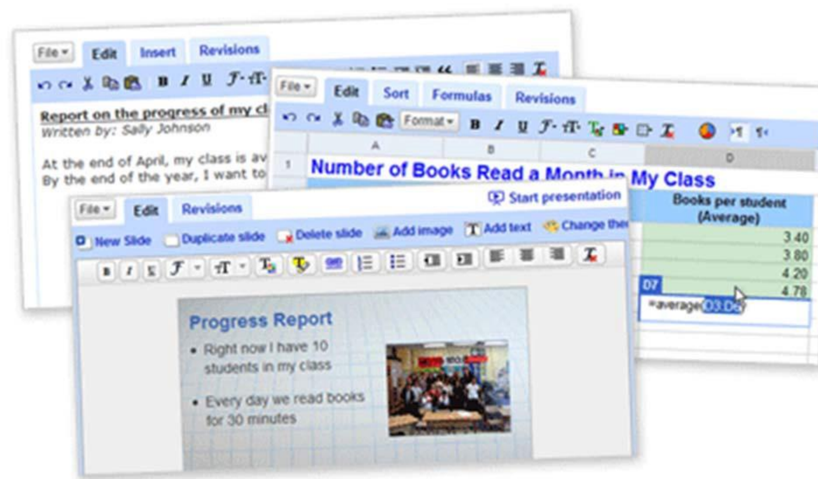
IaaS

- **Infrastructure-as-a-Service**
  - Basic computing resources such as CPU/Memory/Disk, made available to users in the form of Virtual Machine Instances

SaaS

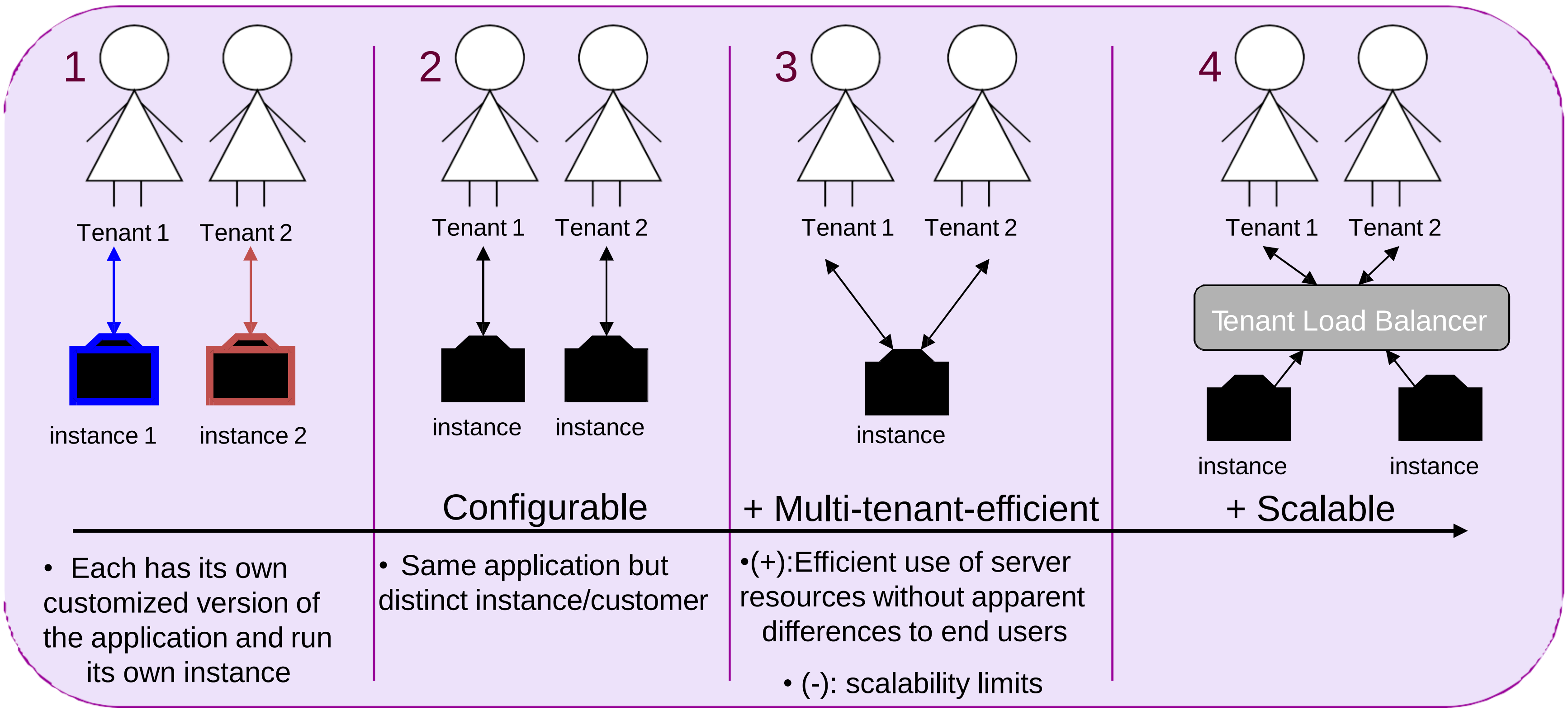
- You are most familiar with this!
- Software is delivered as a service over the Internet, eliminating the need to install and run the application on the customer's own computer
- This simplifies maintenance and support
- Examples: Gmail, YouTube, and Google Docs, among others





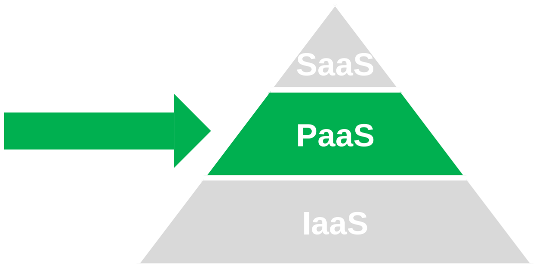
SaaS Maturity Levels

- **Distinguishing attributes:** configurability, multi-tenant efficiency, scalability

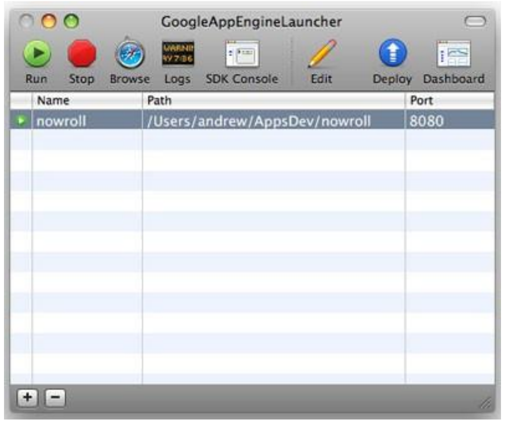


PaaS

- The Cloud provider exposes a set of tools (a platform) which allows users to create SaaS applications



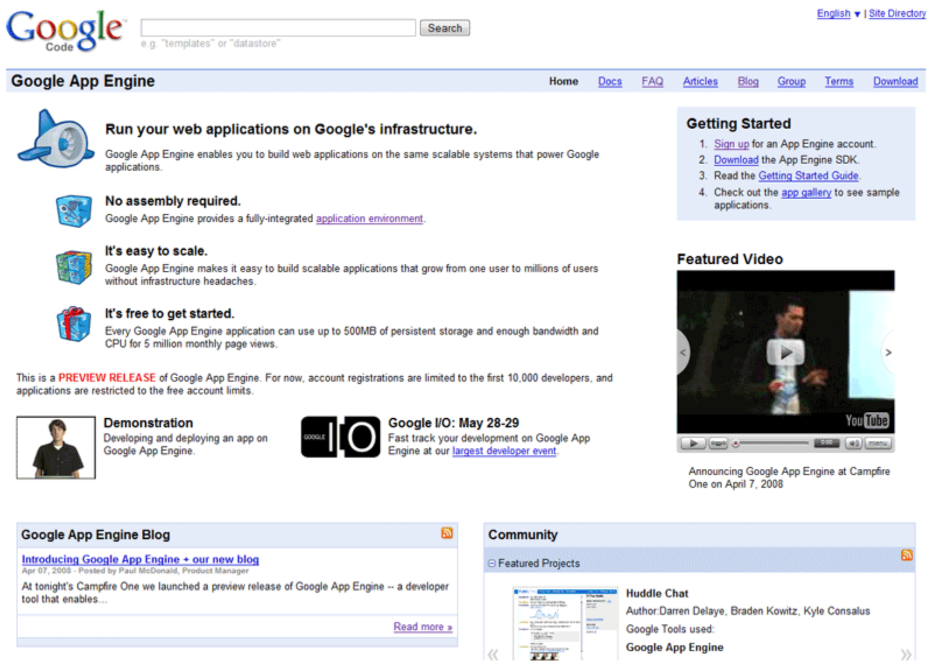
- The SaaS application runs on the provider's infrastructure



- The cloud provider manages the underlying hardware and requirements

PaaS Example I

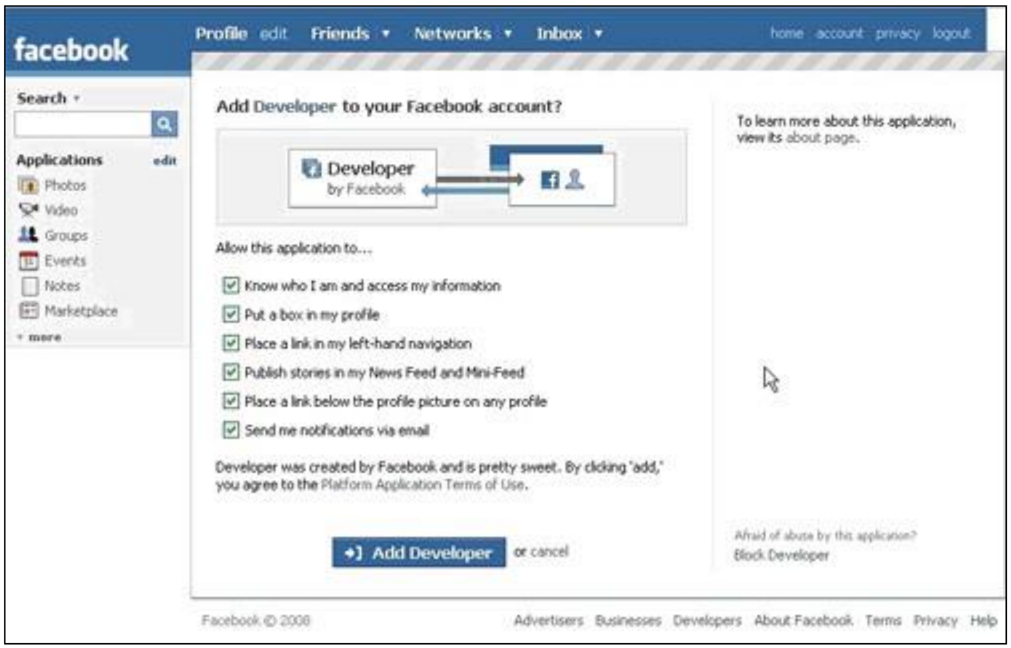
- Google App Engine



Build web applications on Google's Infrastructure

PaaS Example II

- The Facebook Developer Platform

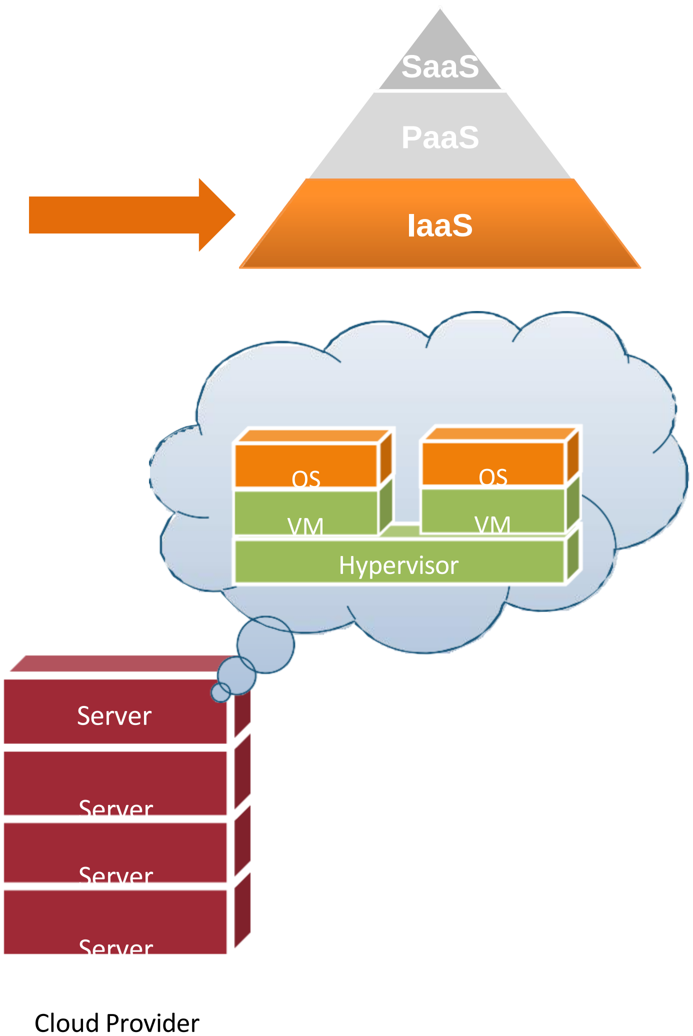


Set of APIs that allow you to create Facebook Applications

IaaS (1/3)

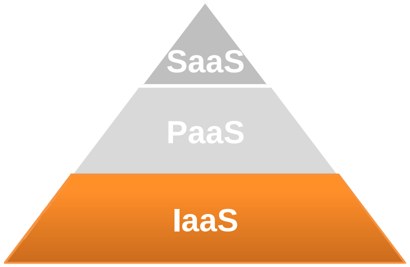
- The cloud provider leases to users Virtual Machine Instances (i.e., computer infrastructure) using the ***virtualization*** technology

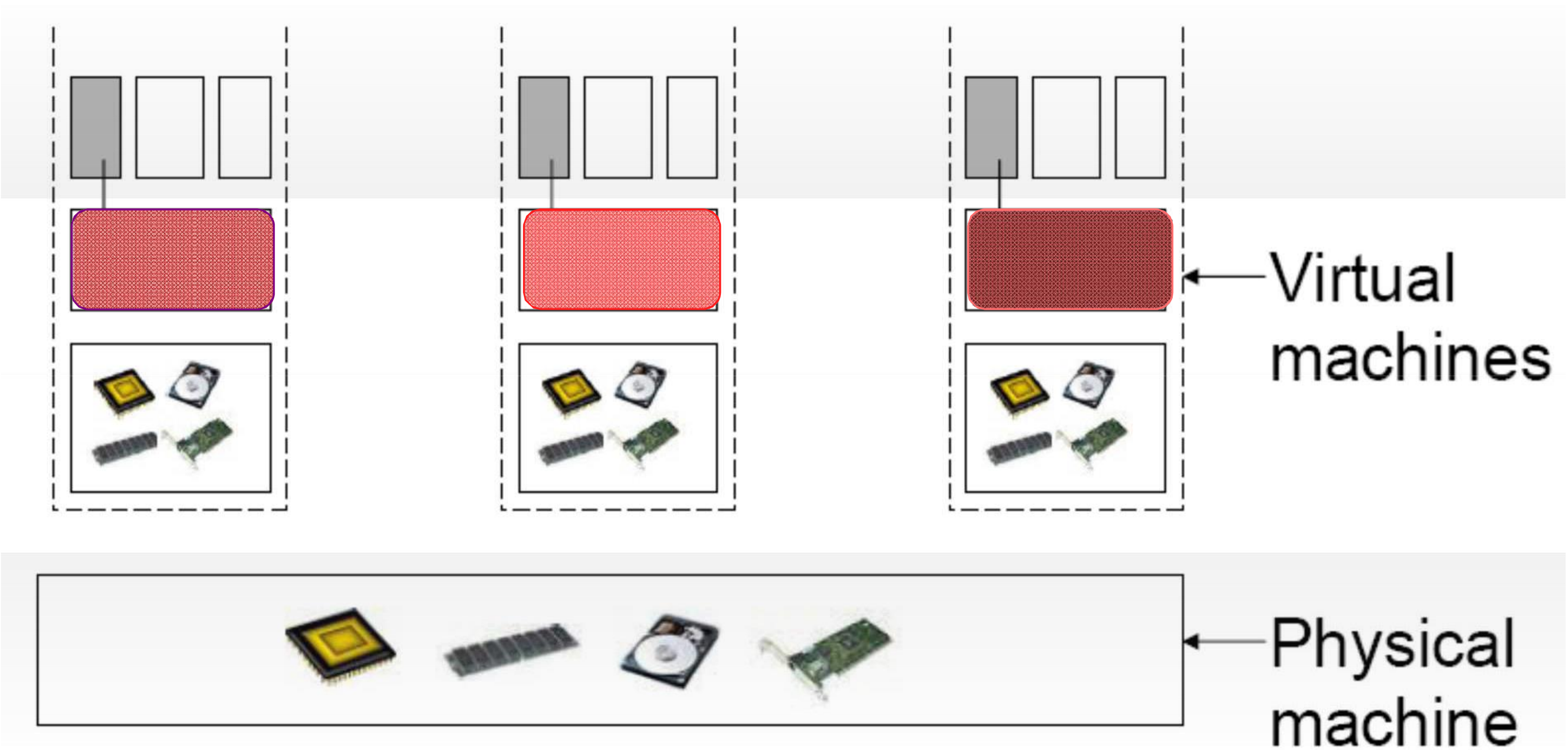
- The user has access to a standard Operating System environment and can install and configure all the layers above it



IaaS (2/3)

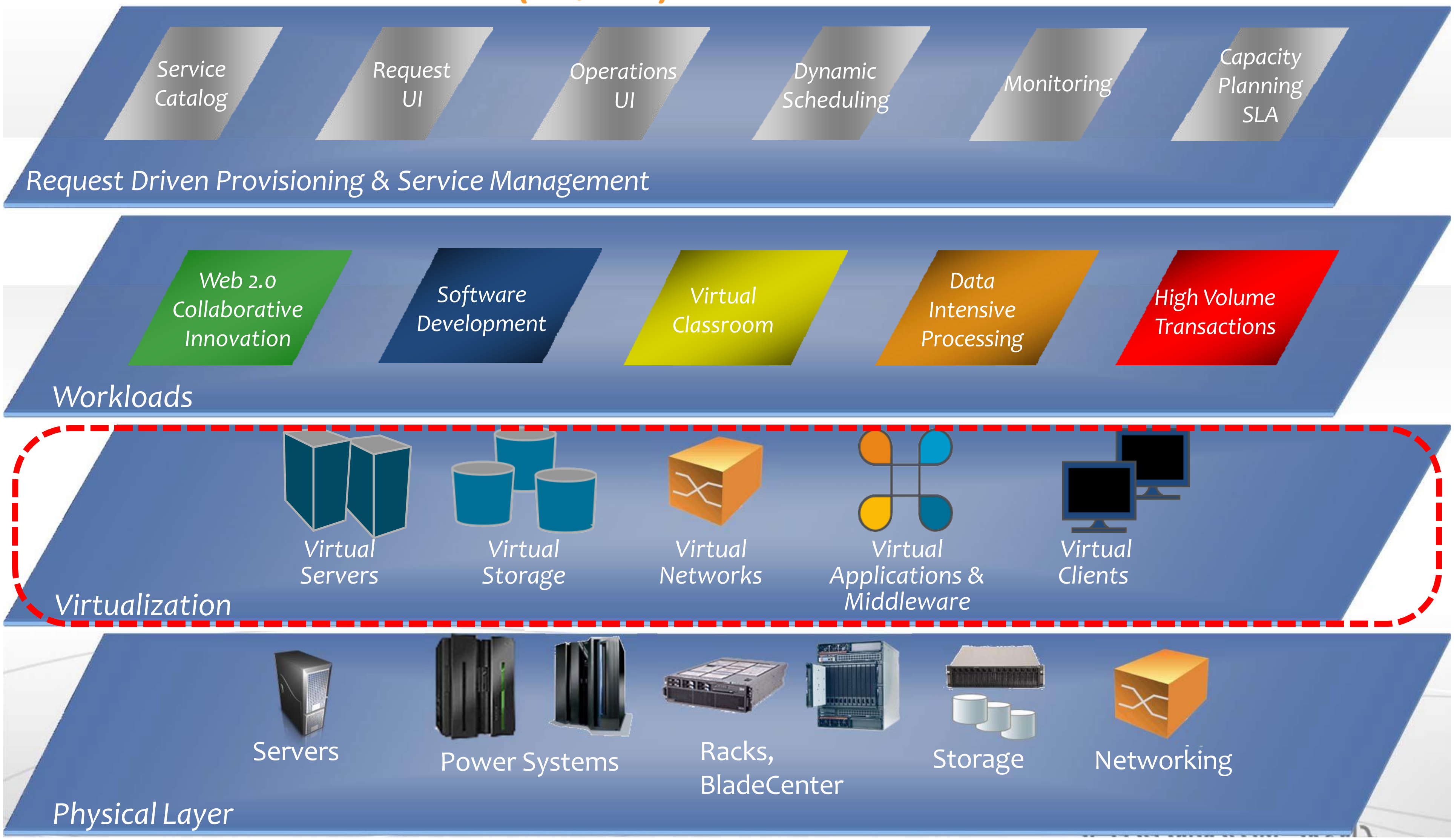
- The virtualization technology is a major enabler of IaaS





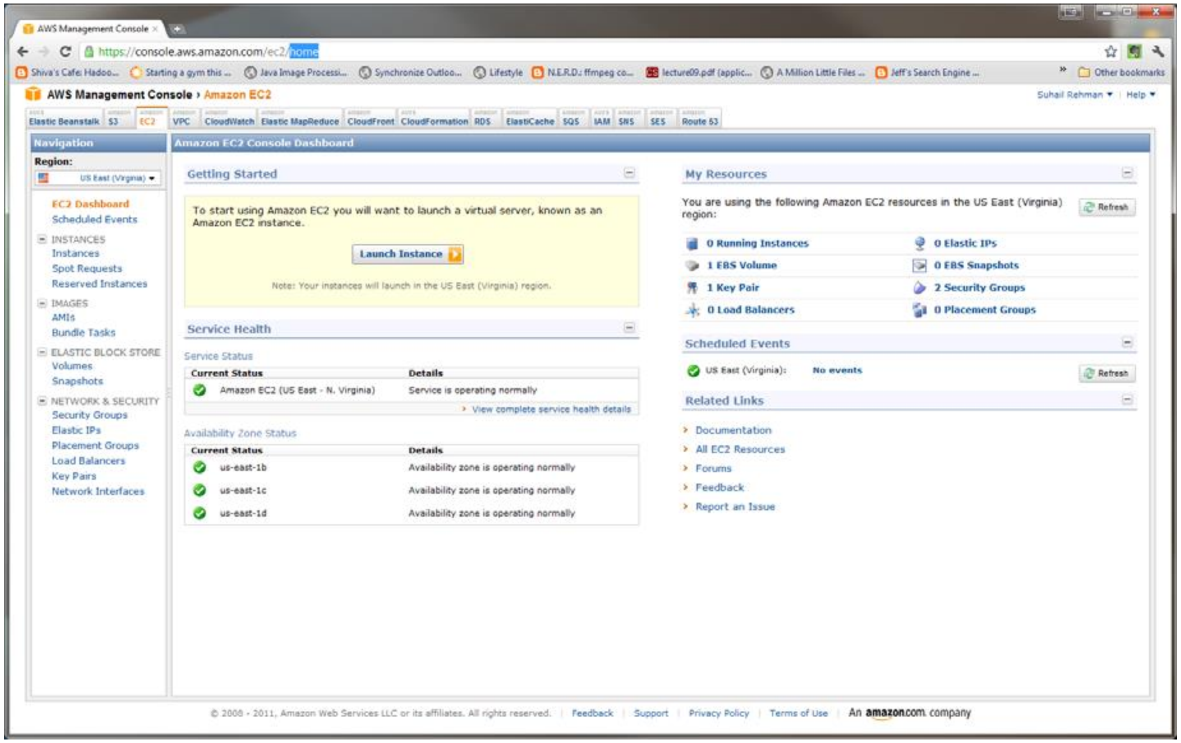
HARDWARE

IaaS (3/3)


Service Catalog
Request UI
Operations UI
Dynamic Scheduling
Monitoring
Capacity Planning SLA
Request Driven Provisioning & Service Management
Web 2.0 Collaborative Innovation
Software Development
Virtual Classroom
Data Intensive Processing
High Volume Transactions
Workloads
Virtual Servers
Virtual Storage
Virtual Networks
Virtual Applications & Middleware
Virtual Clients
Virtualization
Servers
Power Systems
Racks, BladeCenter
Storage
Networking
Physical Layer

IaaS Example

- Amazon Web Service Elastic Compute Cloud (EC2)



Other Service Models

- Hardware-as-a-Service
- Communication-as-a-Service
- XaaS
  - “X” as a Service

Datacenter-as-a-Service

- Increasing Number of Servers
- Manpower, Electricity, Cooling, Security?
  - Management Nightmare
- Why not give it to someone else?

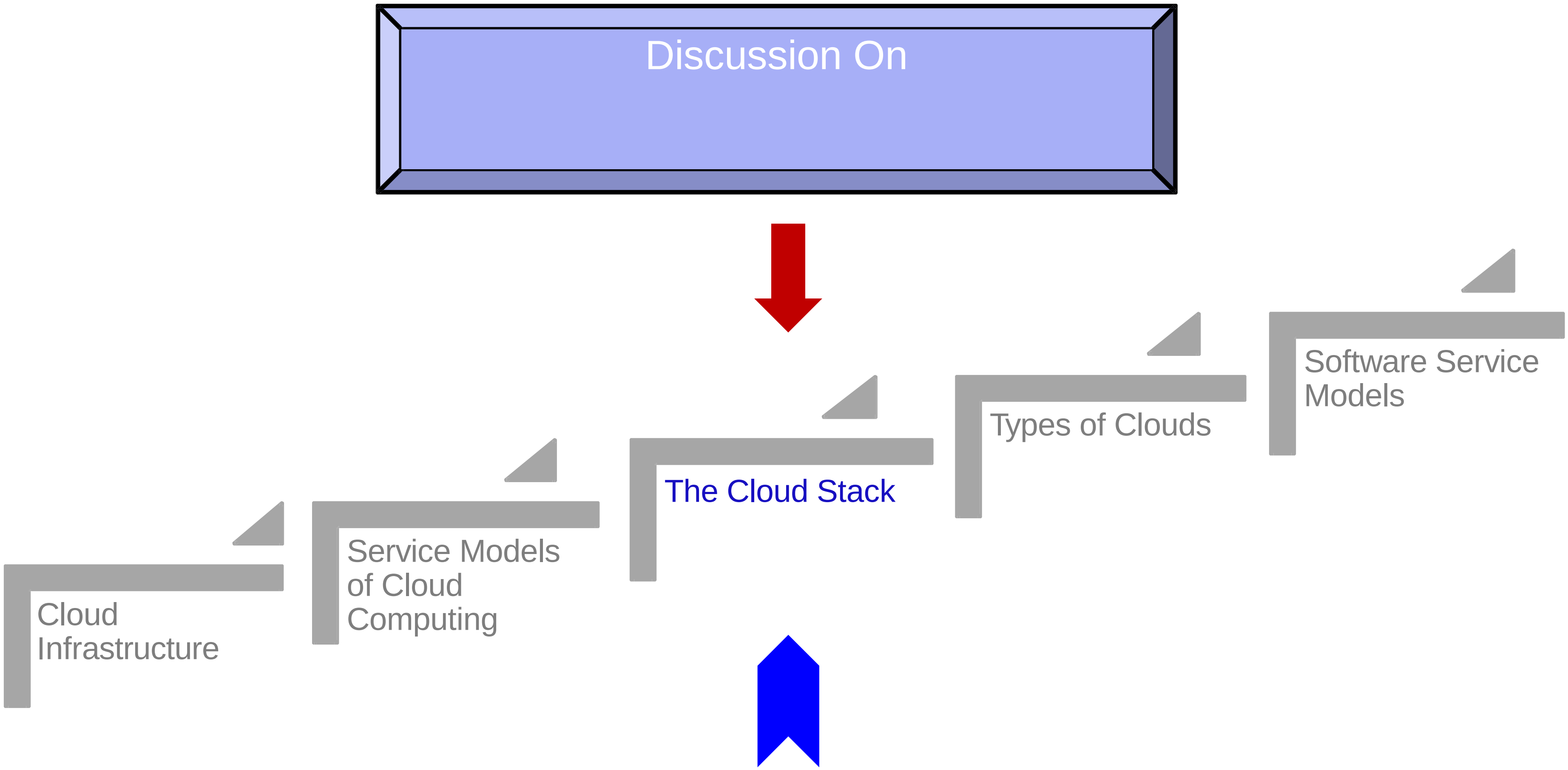
Qatar Airways goes live at Qtel Data Centre | Qatar Telecom (Q-Tel) | AMEinfo.com

http://www.ameinfo.com/260742.html January 5, 2012

Under the agreement, which was signed by Qtel Chairman H.E. Sheikh Abdulla bin Mohammed bin Saud Al-Thani and Qatar Airways Chief Executive Officer Akbar Al Baker, Qatar Airways' core business systems are now operated from the Qtel Data Centre in the country's capital Doha.

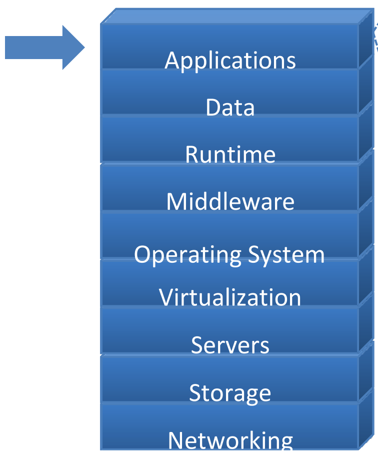
Qtel Chief Executive Officer, Dr. Nasser Marafih, said, "Much has been written about how Qatar Airways changed the game in the airline industry. Now Qatar Airways has enlisted the full range of Qtel's IT and communication solutions to help take its business to an even higher level. Our solutions will harness the power of IT to become an enabler for Qatar Airways business."

Lecture Outline



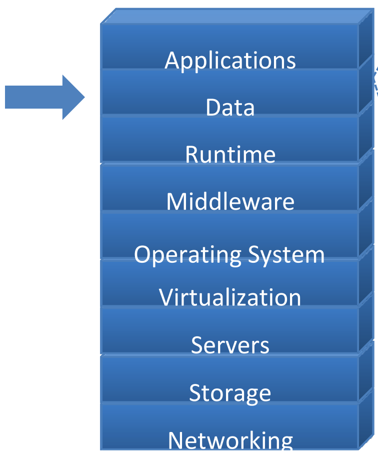
The Cloud Stack

Applications



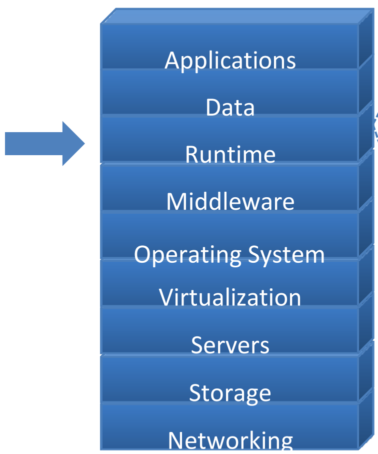
- Cloud applications can range from Web applications to scientific computational jobs

Data



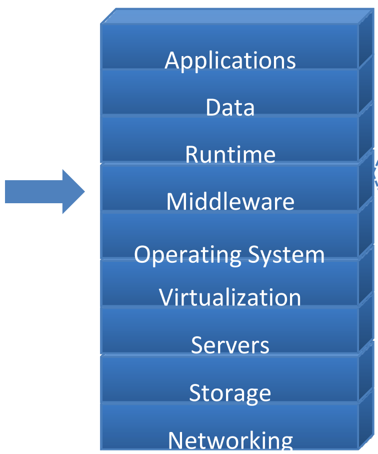
- Data Management
- New generation cloud-specific databases and management systems
- E.g., Hbase, Cassandra, Hive, Pig etc.

Runtime Environment



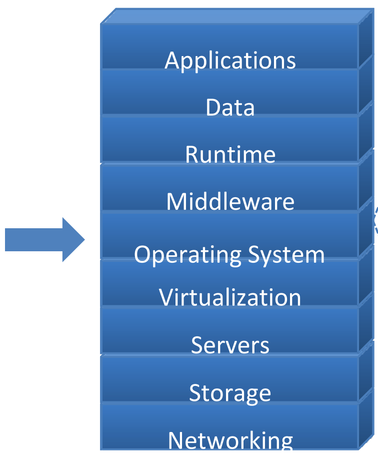
- Runtime platforms to support cloud programming models
- E.g., MPI, MapReduce, Pregel etc.

Middleware for Clouds



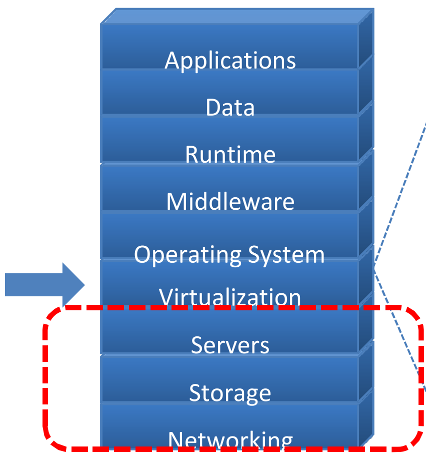
- Management platforms that enable:
  - Resource Management
  - Monitoring
  - Provisioning
  - Identity Management and Security

Operating Systems



- Standard Operating Systems used in Personal Computing
- Packaged with libraries and software for quick deployment and provisioning
- E.g., Amazon Machine Images (AMI) contain OS as well as required software packages as a "snapshot" for instant deployment

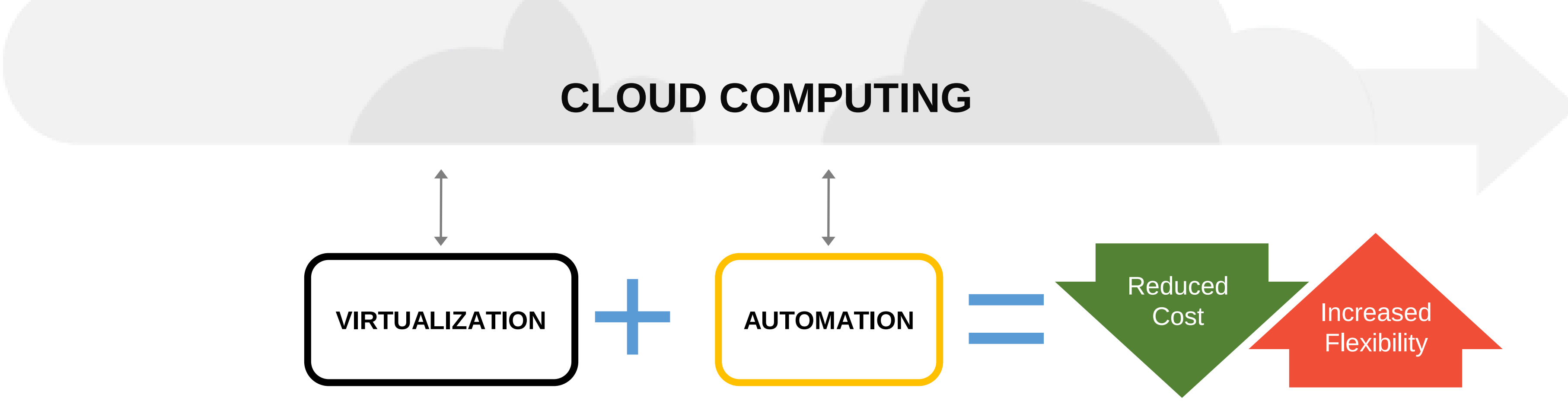
Virtualization



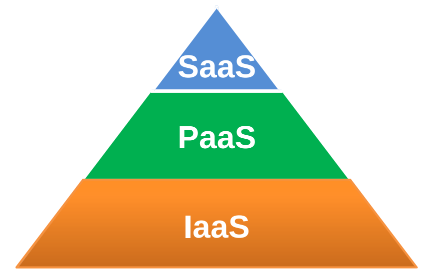
- Key Component
- Resource Virtualization
- Amazon EC2 is based on the Xen virtualization platform

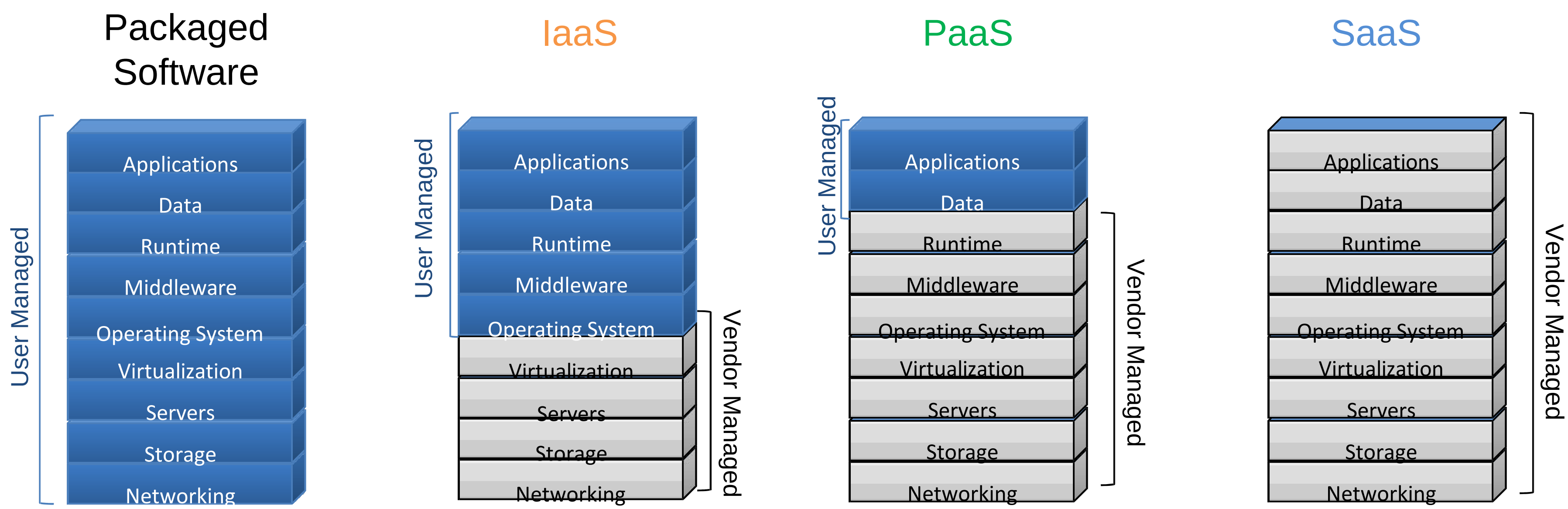
VIRTUALIZATION TECHNOLOGY

- Highly available
- Rapidly deploy new servers
- Easy to deploy
- Reconfigurable while services are running
- Optimizes physical resources by doing more with less

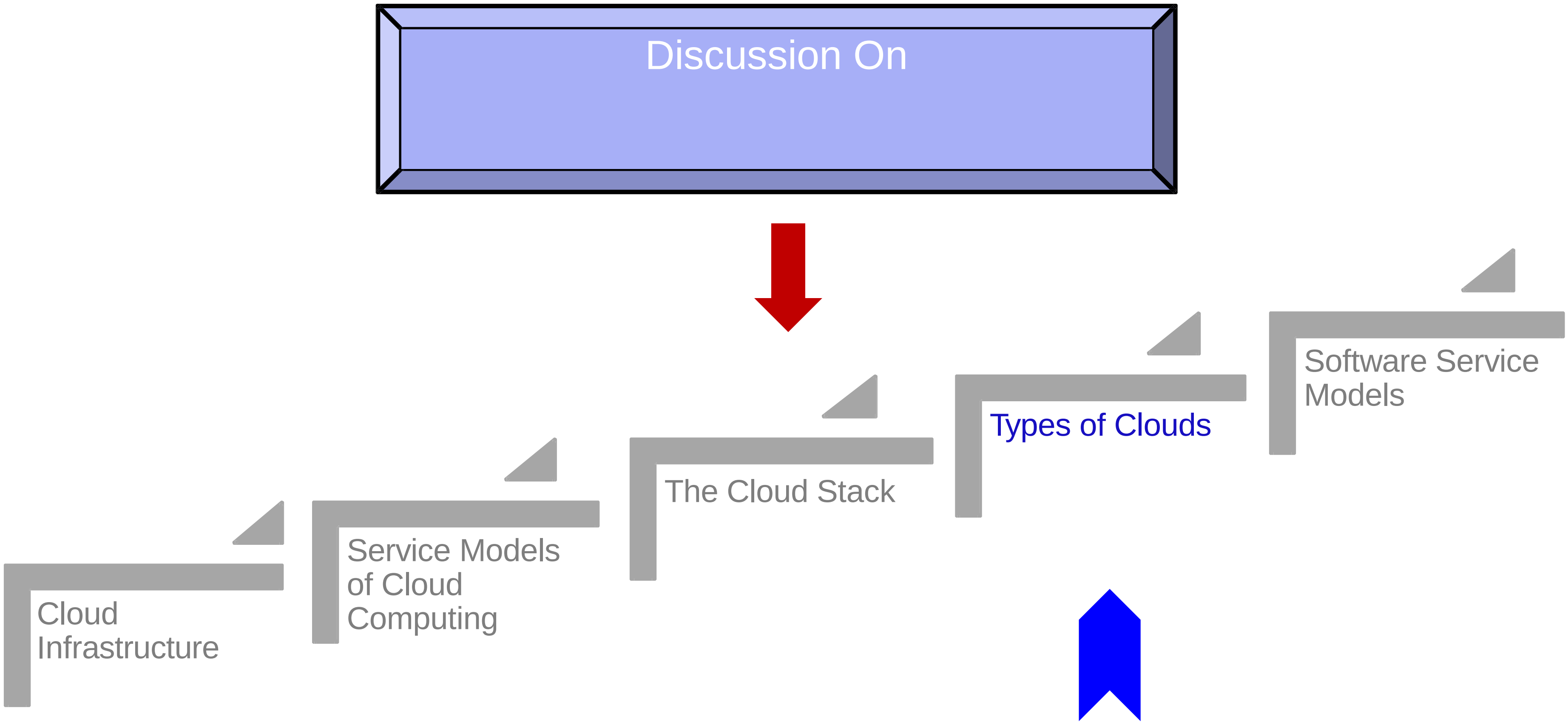


Cloud Service Layers in the Service Levels



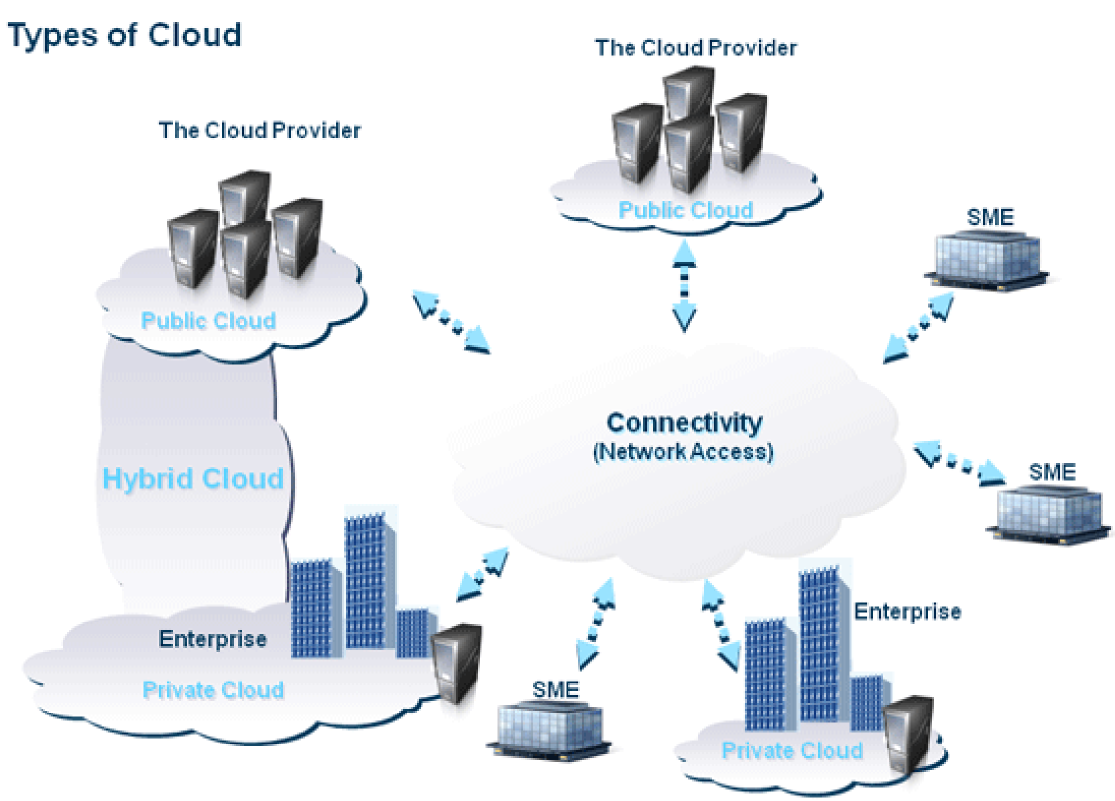


Lecture Outline



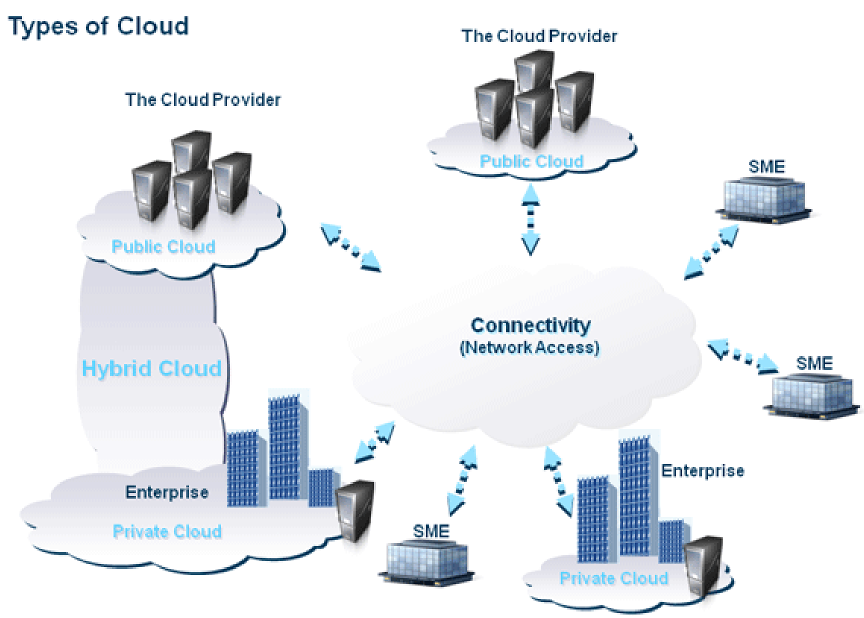
Types of Clouds (1/4)

- Public
- Private
- Hybrid
- Community



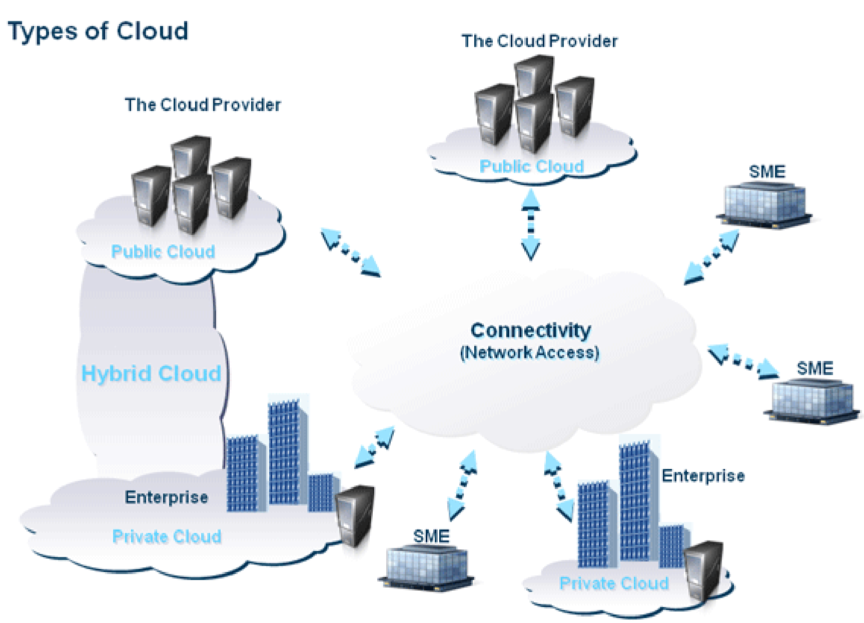
Types of Clouds (2/4)

- Public (external) cloud
  - Open market for on demand computing and IT resources
  - Concerns: Limited SLA, reliability, availability, security, trust and confidence
  - Examples: IBM, Google, Amazon, …



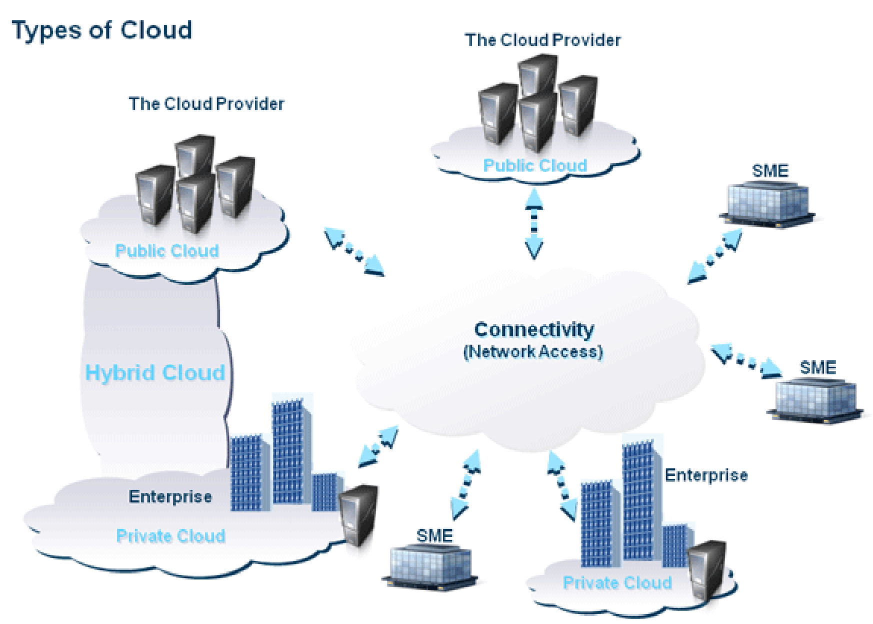
Types of Clouds (3/4)

- Private (Internal) cloud
  - For enterprises/corporations with large scale IT



Types of Clouds (4/4)

- Hybrid cloud
  - Extend the private cloud(s) by connecting it to other external cloud vendors to make use of their available cloud services
- Community



Lecture Outline

Discussion On

Cloud Infrastructure

Service Models of Cloud Computing

The Cloud Stack

Types of Clouds

Software Service Models

Economics of Cloud Computing

- Evolution of Software Service Models
- What is the Value Proposition for Cloud Computing?
- How did Cloud Computing emerge from business / industry rather than from Academia?

Cost of Information Technology

- When you are using IT there are three primary costs associated with it:
  - Software Cost (Media + License cost/user)
  - Support Cost (Vendor Support, Updates and Patches etc.)
  - Management Cost (IT Infrastructure costs, Manpower, etc.)

Traditional Model

- Classical Model
- Software provider develops software and charges a license fee per user for the client
- The provider may charge a support fee /user
- The management of the software is the clients responsibility
  - Up to 4x the cost of the actual software per year!
  - Infrastructure, Manpower, software maintenance
- Traditional Software – Oracle etc.

Open Source Model

- “Free” Model
- Software provider packages Open Source Software and provides it at little or no cost to the client

Outsourcing Model

- Primary cost of Software Management is in Manpower
- Why not delegate the management of software to a country with cheaper labor costs
  - India, China etc.
- Outsource the management of software for a flat fee – keep IT management costs under control

Software as a Service Cloud Computing

- Develop Web Application
- Offer to customers over Internet
- Amortize Management and Support costs over many clients

Questions?